🧙 Spell Arena

Real-Time Multiplayer Gesture-Based Combat

Project Plan Presentation

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Agenda

1. Problem Domain & Motivation
2. Features & Requirements
3. System Architecture
4. Core Networked Components
5. Design Patterns
6. Testing Strategy
7. Demo & Links

Problem Domain

The Challenge

Building **multiplayer functionality** into an existing single-player game presents unique challenges:

- 🔄 **State Synchronization** across all clients
- ⏱️ **Latency Compensation** for responsive gameplay
- 🎯 **Real-time Gesture Recognition** with spell casting
- 🌐 **Network Reliability** and disconnect handling

Why It Matters

Challenge	Impact
Synchronized game state	All players see the same reality
Consistent spell projectiles	Fair competitive gameplay
Steam integration	Reliable transport & authentication
P2P architecture	Reduced server costs

The Solution

Wizard Hands Multiplayer

Extends the single-player gesture-based combat into a **networked experience**:

- **Netcode for GameObjects** - State synchronization
- **Facepunch.Steamworks** - Steam transport layer
- **Peer-to-Peer** - Friend invites, lobby creation, NAT traversal

Players duel using **hand gestures** to cast spells! 🪄

Feature 1: Steam Lobby & Matchmaking

ID	User Story	Acceptance Test
R1.1	Create a Steam lobby	Verify lobby appears in Steam overlay
R1.2	Join friend's lobby via invite	Both players appear in lobby
R1.3	See connected players before start	Player list updates real-time
R1.4	Host starts match when ready	All clients transition to game

Feature 2: Player Synchronization

ID	User Story	Acceptance Test
R2.1	See other players' positions	Position updates within 100ms
R2.2	See synchronized animations	Animation plays on all clients
R2.3	See other players' health	Health bar updates for all

Feature 3: Networked Spell Casting

ID	User Story	Acceptance Test
R3.1	See gesture inputs displayed	Gesture visible on remote clients
R3.2	See spell projectiles	Projectile spawns on all clients
R3.3	See spell effects (shields, auras)	Shield visual on all clients

Feature 4: Combat & Damage Sync

ID	User Story	Acceptance Test
R4.1	Deal damage to remote players	Damage applied on both clients
R4.2	Receive damage from remote spells	Health decreases with feedback
R4.3	Authoritative damage calculation	Both clients agree on health

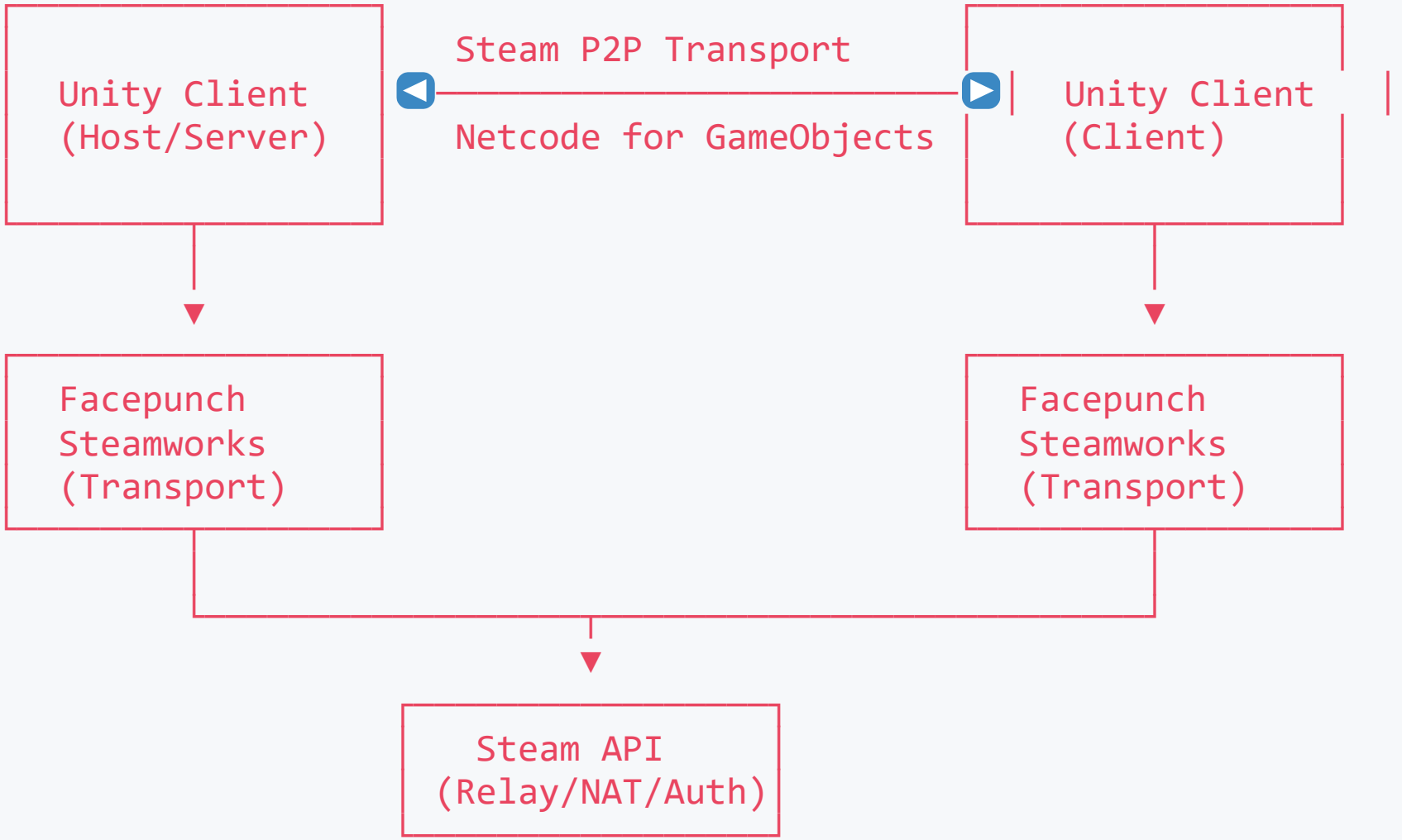
Feature 5: Game State & Win Condition

ID	User Story	Acceptance Test
R5.1	See when opponent is defeated	Defeat state shown on all clients
R5.2	See win/lose screen at match end	Correct winner displayed
R5.3	Rematch or return to lobby	Both players restart correctly

Feature 6: Network Resilience

ID	User Story	Acceptance Test
R6.1	See when opponent disconnects	Disconnect notification shown
R6.2	Return to lobby after disconnect	Player can return to menu
R6.3	Handle host migration	New host assigned gracefully

System Architecture



Core Networked Components

Component	Description	Key NetworkVariables/RPCs
NetworkedPlayer	Synchronized player state	position, rotation, health
NetworkedSpellCaster	Spell casting across network	ServerRpc: CastSpell
NetworkedProjectile	Synchronized projectiles	NetworkTransform, damage
NetworkedHealth	Player health with authority	currentHealth, isDead
LobbyManager	Steam lobby management	lobbyId, playerList
SteamNetworkTransport	Facepunch transport layer	SteamId, connection

Network Architecture Decisions

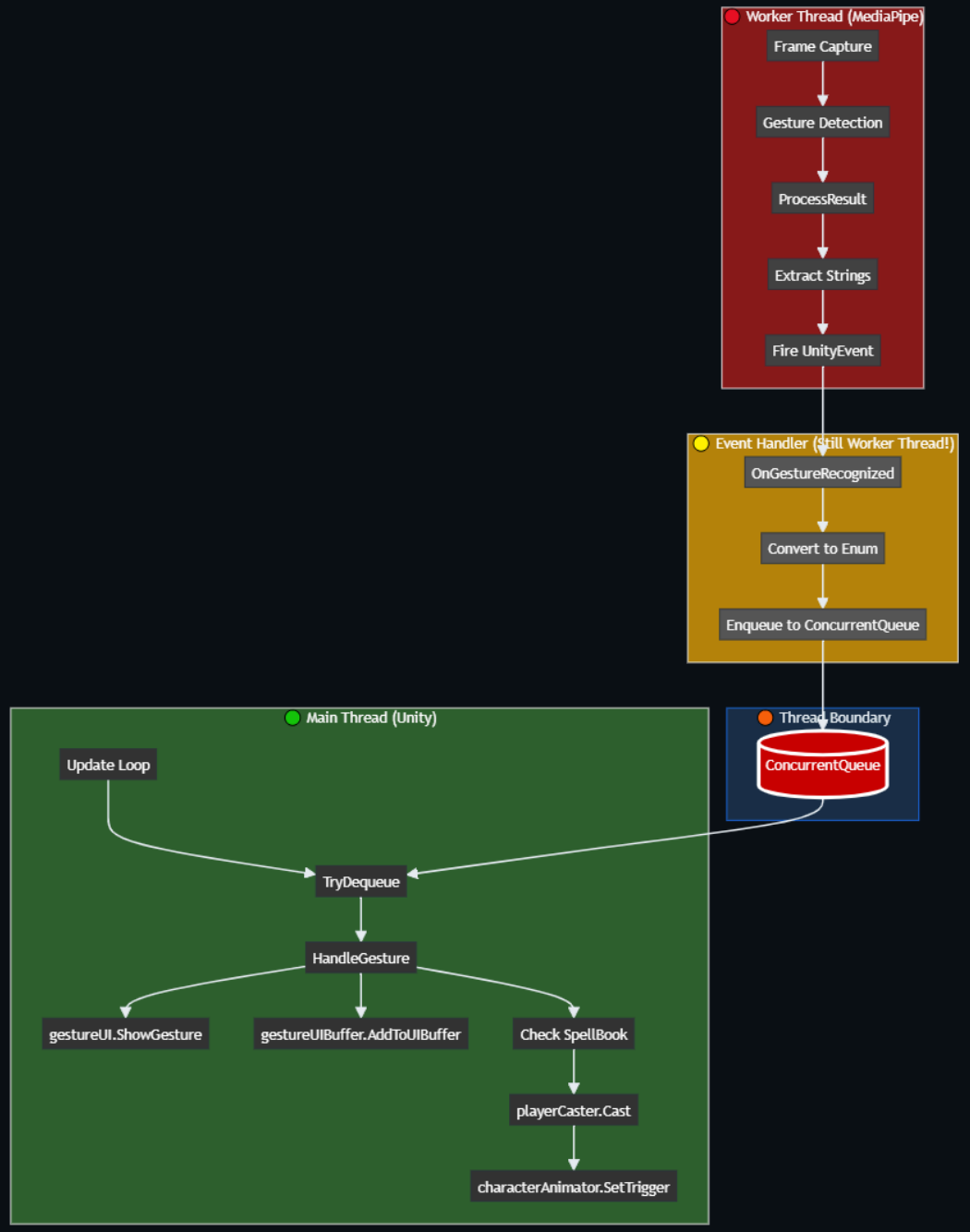
Decision	Choice	Rationale
Network Topology	Host-Client (P2P)	No dedicated server; Steam handles NAT
State Sync	Server-Authoritative	Host validates damage and state
Transport	Facepunch.Steamworks	Steam relay, friend integration
Projectile Spawning	Server-spawned	Prevents duplicates & cheating
Health/Damage	Server-authoritative	Host calculates & broadcasts

Game Flow

```
graph TD
    A[Startup] --> B[Load Assets]
    B --> C[Main Menu]
    C -->|Start Game| D[Load Game Scene]
    D --> E[Initialize Camera Feed]
    E --> F[Start Gesture Recognition Loop]
    F --> G{Hands Detected?}
    G -->|Two Hands| I[Track Gesture Sequence]
    I --> M[Submit Sequence]
    M --> N{Spell Match?}
    N -->|Yes| O[Cast Spell]
    O --> P[Reset Buffer]
    P --> F
```

Threading Model

- **Worker Thread**: MediaPipe processing
- **Thread Boundary**: ConcurrentQueue buffer
- **Main Thread**: Unity API calls

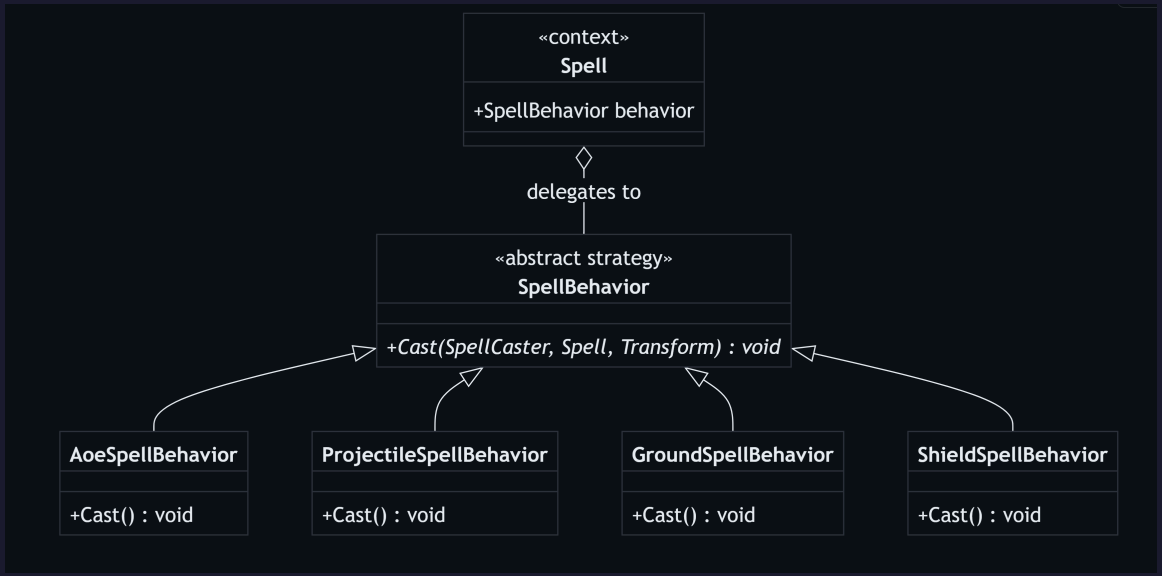


Design Pattern: Strategy ⭐

Location: `Spell.cs` holds SpellBehavior reference

Benefits:

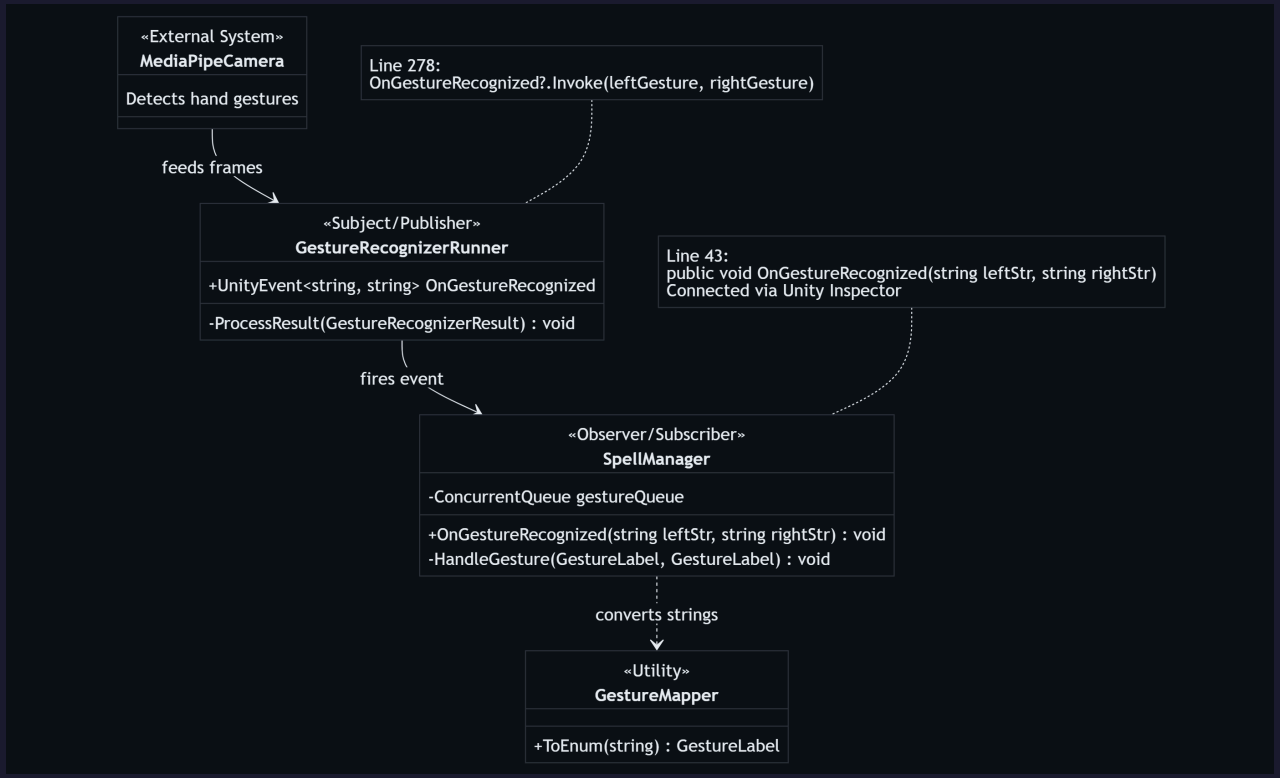
- Add new spell types easily
- Isolated & testable behaviors
- Runtime switching possible



Design Pattern: Observer

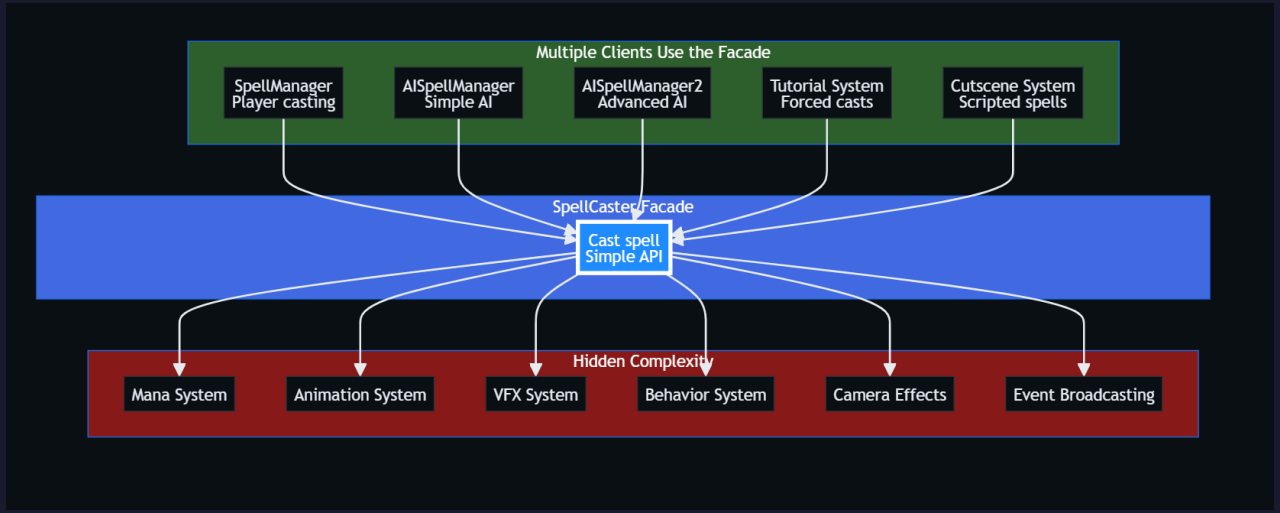
GestureRecognizerRunner publishes gesture events

SpellManager subscribes and handles gestures



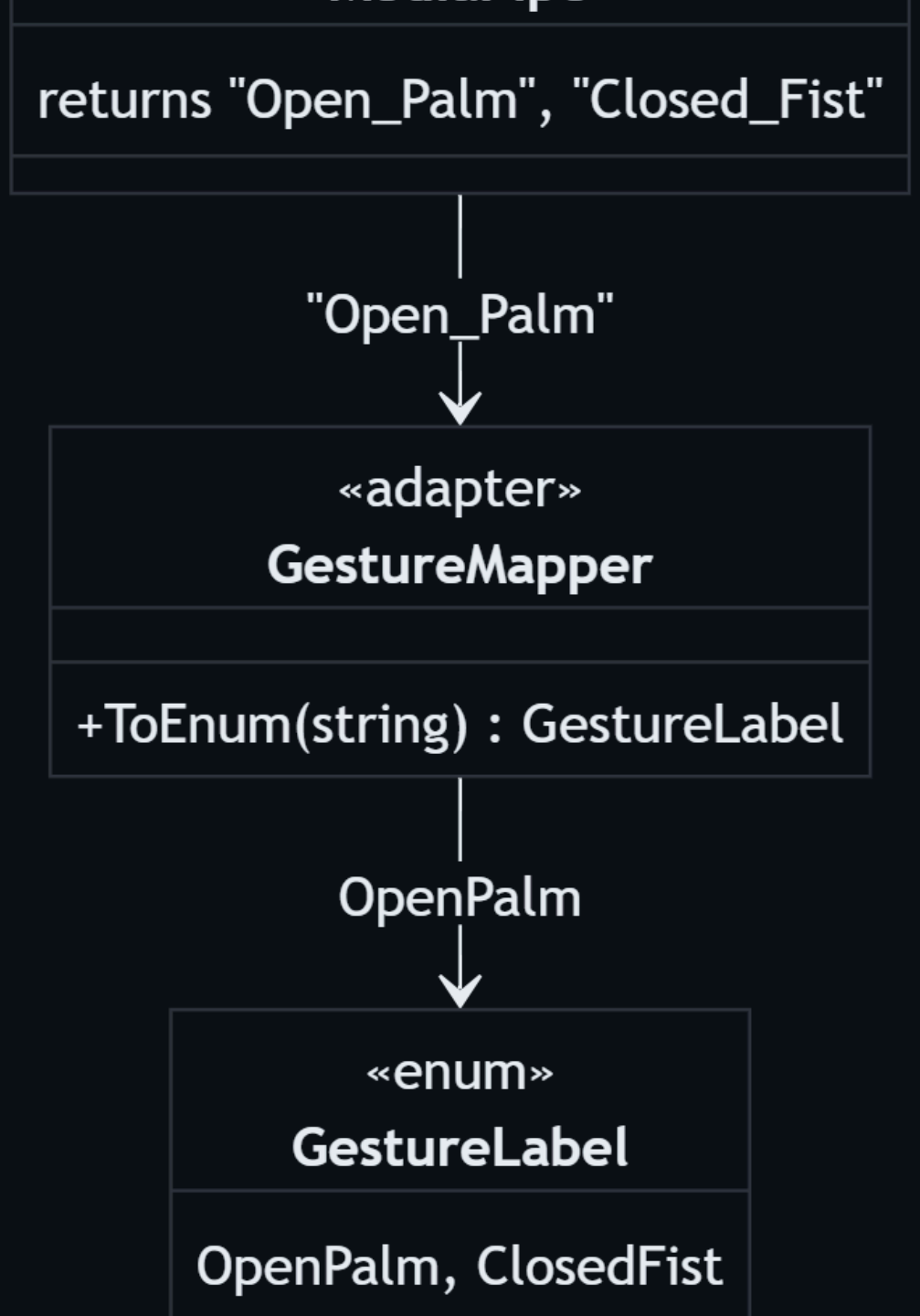
Design Pattern: Facade

SpellCaster provides simplified interface to complex spell-casting subsystem



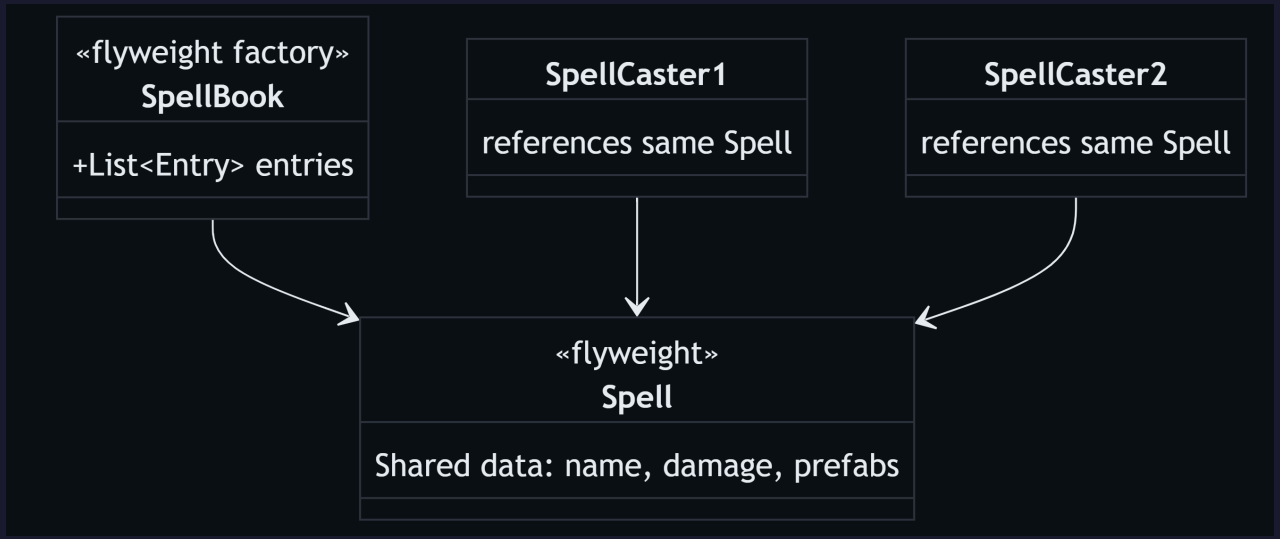
Design Pattern: Adapter

GestureMapper adapts MediaPipe strings to internal enums



Design Pattern: Flyweight

ScriptableObjects share data across multiple instances



Additional Patterns

Component Pattern (Unity)

- SpellCaster ↔️ Animator
- SpellCaster ↔️ ShieldComponent
- ProjectileBase ↔️ Rigidbody

Manager/Service Pattern

- **SpellManager** - gesture input → spell casting
- **AISpellManager** - AI spell orchestration
- **CameraEffects** - camera shake service

Clean Architecture Separation

Layer	Components
Data	Spell, SpellBook (ScriptableObjects)
Behavior	SpellBehavior (Strategy Pattern)
Execution	SpellCaster, AISpellManager
Input	SpellManager, GestureRecognizerRunner
UI	GestureUI, GestureUIBuffer, EnemyGestureDisplay

Test Strategy

Testing Approach

Type	Coverage
Unit Tests	NetworkVariable serialization, damage calculation
Integration Tests	Lobby creation/joining, player spawning, RPC delivery
Acceptance Tests	Full multiplayer matches with gesture input

Burndown Metrics

Metric	Count
Features	6
Requirements	19
Tests	19

All requirements have direct acceptance tests defined ✅

Project Links

Resource	Link
GitHub Repository	github.com/jp-loran/wizards
Source Code	`./code/`
Documentation	`./docs/`
Demo Download	Google Drive

Developer

Name	Role
Joe Ampfer	Developer

Course: ASE 330 - Human Computer Interaction

🎮 Demo Time!

Questions?

Try the demo: Download for Windows

Thank You! 🧙

Spell Arena - Real-Time Multiplayer Gesture-Based Combat

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